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chapter

23 Hormonal Regulation and Integration of Mammalian Metabolism

1. **Peptide Hormone Activity** Explain how two peptide hormones as structurally similar as oxytocin and vasopressin can have such different effects (see Fig. 23–10).

   Answer Oxytocin and vasopressin are recognized by different receptors, typically found in different cell types. These receptors are coupled to different downstream effects in their target cells.

2. **ATP and Phosphocreatine as Sources of Energy for Muscle** During muscle contraction, the concentration of phosphocreatine in skeletal muscle drops while the concentration of ATP remains fairly constant. However, in a classic experiment, Robert Davies found that if he first treated muscle with 1-fluoro-2,4-dinitrobenzene (p. 94), the concentration of ATP declined rapidly while the concentration of phosphocreatine remained unchanged during a series of contractions. Suggest an explanation.

   Answer Muscle contraction results in a net hydrolysis of ATP. Although the amount of ATP in muscle is very small, the supply can be rapidly replenished by phosphoryl group transfer from the phosphocreatine reservoir, catalyzed by creatine kinase:

   $$\text{Phosphocreatine} + \text{ADP} \rightleftharpoons \text{creatine} + \text{ADP}$$

   Because this reaction is rapid relative to the use of ATP by muscle, [ATP] remains in a steady state. The effect of pretreatment with fluoro-2,4-dinitrobenzene suggests that this is an effective inhibitor of creatine kinase. Under working conditions, the small amount of muscle ATP is quickly depleted and cannot be replenished.

3. **Metabolism of Glutamate in the Brain** Brain tissue takes up glutamate from the blood, transforms it into glutamine, then releases it into the blood. What is accomplished by this metabolic conversion? How does it take place? The amount of glutamine produced in the brain can actually exceed the amount of glutamate entering from the blood. How does this extra glutamine arise? (Hint: you may want to review amino acid catabolism in Chapter 18; recall that NH_3 is very toxic to the brain.)

   Answer Ammonia is very toxic to nervous tissue, especially the brain. Excess NH_3 in brain cells is removed by the transformation of glutamate to glutamine, catalyzed by glutamine synthetase. Glutamine is then exported to the blood and travels to the liver, where it is transformed to urea.

   The additional glutamine in the brain arises from the action of aminotransferases that transfer amino groups from amino acids to α-ketoglutarate (a citric acid cycle intermediate), forming glutamate, which is then converted to glutamine.

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4. **Proteins as Fuel during Fasting** When muscle proteins are catabolized in skeletal muscle during a fast, what are the fates of the amino acids?

   Answer Glucogenic amino acids are used to make glucose for the brain; others are oxidized in mitochondria via the citric acid cycle.

5. **Absence of Glycerol Kinase in Adipose Tissue** Glycerol 3-phosphate is required for the biosynthesis of triacylglycerols. Adipocytes, specialized for the synthesis and degradation of triacylglycerols, cannot use glycerol directly because they lack glycerol kinase, which catalyzes the reaction

   $$\text{Glycerol} + \text{ATP} \longrightarrow \text{glycerol 3-phosphate} + \text{ADP}$$

   How does adipose tissue obtain the glycerol 3-phosphate necessary for triacylglycerol synthesis?

   Answer Adipose tissue converts glucose to the glycolytic intermediate dihydroxyacetone phosphate, which is reduced to glycerol 3-phosphate by the NADH-requiring enzyme glycerol 3-phosphate dehydrogenase:

   $$\text{Glucose} \xrightarrow{\text{glycolysis}} \text{dihydroxyacetone phosphate}$$
   $$\text{Dihydroxyacetone phosphate} + \text{NADH} + \text{H}^+ \rightleftharpoons \text{glycerol 3-phosphate} + \text{NAD}^+$$

6. **Oxygen Consumption during Exercise** A sedentary adult consumes about 0.05 L of O_2 in 10 seconds. A sprinter, running a 100 m race, consumes about 1 L of O_2 in 10 seconds. After finishing the race, the sprinter continues to breathe at an elevated (but declining) rate for some minutes, consuming an extra 4 L of O_2 above the amount consumed by the sedentary individual.
   (a) Why does the need for O_2 increase dramatically during the sprint?
   (b) Why does the demand for O_2 remain high after the sprint is completed?

   Answer
   (a) Increased muscle activity raises the demand for ATP, which is met by increased activity of the citric acid cycle enzymes and increased flow of electrons through the electron-transfer chain. This results in greater O_2 consumption.
   (b) During the sprint, muscle transforms some glycogen to lactate (anaerobic glycolysis). After the sprint, lactate is transported to the liver, where it is converted back to glucose and glycogen. This process requires ATP and thus requires O_2 consumption above the resting rate.

7. **Thiamine Deficiency and Brain Function** Individuals with thiamine deficiency show some characteristic neurological signs and symptoms, including loss of reflexes, anxiety, and mental confusion. Why might thiamine deficiency be manifested by changes in brain function?

   Answer Glucose is the primary fuel of the brain, and the brain is particularly sensitive to any change in the availability of glucose for energy production. A key reaction in glucose catabolism is the thiamine pyrophosphate–dependent oxidative decarboxylation of pyruvate to acetyl-CoA. Thus, a thiamine deficiency reduces the rate of glucose catabolism.

8. **Potency of Hormones** Under normal conditions, the human adrenal medulla secretes epinephrine ($C_9H_{13}NO_3$) at a rate sufficient to maintain a concentration of 10^{-10} M in circulating blood. To appreciate what that concentration means, calculate the diameter of a round swimming pool, with a water depth of 2.0 m, that would be needed to dissolve 1.0 g (about 1 teaspoon) of epinephrine to a concentration equal to that in blood.

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Answer The concentration of epinephrine in blood is 10^{-10} M = 10^{-10} mol/L. The molecular weight of epinephrine is 183.21 g/mol. Thus, the mass of epinephrine (in grams) needed for a concentration of 10^{-10} mol/L is

$$(10^{-10}\ \text{mol/L})(183.21\ \text{g/mol}) = 1.8 \times 10^{-8}\ \text{g/L}$$

If ν is the volume in liters needed to dissolve 1 g of epinephrine to this concentration,

$$1.8 \times 10^{-8}\ \text{g/L} = 1\ \text{g}/\nu$$
$$\nu = (1\ \text{g})/(1.8 \times 10^{-8}\ \text{g/L}) = 5.5 \times 10^{7}\ \text{L} = 5.5 \times 10^{10}\ \text{cm}^3$$

If r is the radius of the pool, and h is the height of the water, the volume of water is $\nu = \pi r^2 h$ (volume of a cylinder). Rearranging to solve for r^2:

$$r^2 = \nu/\pi h$$

and substituting for ν, π, and h:

$$r^2 = (5.5 \times 10^{10}\ \text{cm}^3)/3.14(200\ \text{cm}) = 8.8 \times 10^{7}\ \text{cm}^2$$

Taking the square root:

$$r = 9{,}400\ \text{cm} = 94\ \text{m}$$

The diameter of the pool = $2r$ = 190 m. For comparison, an Olympic-size swimming pool is 50 m long.

9. **Regulation of Hormone Levels in the Blood** The half-life of most hormones in the blood is relatively short. For example, when radioactively labeled insulin is injected into an animal, half of the labeled hormone disappears from the blood within 30 min.
   (a) What is the importance of the relatively rapid inactivation of circulating hormones?
   (b) In view of this rapid inactivation, how is the level of circulating hormone kept constant under normal conditions?
   (c) In what ways can the organism make rapid changes in the level of a circulating hormone?

   Answer
   (a) Inactivation provides a rapid means to change hormone concentrations.
   (b) A constant insulin level is maintained by equal rates of synthesis and degradation.
   (c) Other means of varying hormone concentration include changes in rate of release from storage, rate of transport, and rate of conversion from prohormone to active hormone.

10. **Water-Soluble versus Lipid-Soluble Hormones** On the basis of their physical properties, hormones fall into one of two categories: those that are very soluble in water but relatively insoluble in lipids (e.g., epinephrine) and those that are relatively insoluble in water but highly soluble in lipids (e.g., steroid hormones). In their role as regulators of cellular activity, most water-soluble hormones do not enter their target cells. The lipid-soluble hormones, by contrast, do enter their target cells and ultimately act in the nucleus. What is the correlation between solubility, the location of receptors, and the mode of action of these two classes of hormones?

    Answer Because of their low solubility in lipids, water-soluble hormones cannot penetrate the plasma membrane; they bind to receptors on the outer surface of the cell. In the case of epinephrine, this receptor is an enzyme that catalyzes the formation of a second messenger (cAMP) inside the cell. In contrast, lipid-soluble hormones readily penetrate the hydrophobic core of the plasma membrane. Once inside the cell, they can act on their target molecules or receptors directly.

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11. Metabolic Differences between Muscle and Liver in a "Fight or Flight" Situation When an animal confronts a "fight or flight" situation, the release of epinephrine promotes glycogen breakdown in the liver, heart, and skeletal muscle. The end product of glycogen breakdown in the liver is glucose; the end product in skeletal muscle is pyruvate.

(a) What is the reason for the different products of glycogen breakdown in the two tissues?

(b) What is the advantage to an animal that must fight or flee of these specific glycogen breakdown routes?

Answer

(a) Glycogen breakdown in hepatocytes in response to epinephrine produces glucose 6-phosphate, which can be dephosphorylated by glucose 6-phosphatase; the glucose is then exported into the bloodstream. Heart and skeletal muscle lack glucose 6-phosphatase. Any glucose 6-phosphate produced in these tissues enters the glycolytic pathway and is metabolized to pyruvate. Under O_2-deficient conditions (actively contracting muscle), pyruvate is converted to lactate to regenerate NAD^+ for continued glycolysis.

(b) In a "fight or flight" situation, the concentration of glycolytic precursors in muscle tissue needs to be high in preparation for activity. Phosphorylated intermediates, such as glucose 6-phosphate, cannot escape from the cell because the membrane is not permeable to charged species; and glucose 6-phosphate is not exported by the glucose transporter. Thus, glucose 6-phosphate is retained in the cell and enters glycolysis. The liver, by contrast, does export glucose, which is necessary to maintain the blood glucose level. The glucose is transported in the bloodstream for uptake and use by muscle.



12. Excessive Amounts of Insulin Secretion: Hyperinsulinism Certain malignant tumors of the pancreas cause excessive production of insulin by the β cells. Affected individuals exhibit shaking and trembling, weakness and fatigue, sweating, and hunger.

(a) What is the effect of hyperinsulinism on the metabolism of carbohydrates, amino acids, and lipids by the liver?

(b) What are the causes of the observed symptoms? Suggest why this condition, if prolonged, leads to brain damage.

Answer

(a) Overproduction of insulin leads to excessive uptake of blood glucose by the liver and conversion to glycogen, leading to hypoglycemia. In addition, a general shutdown of amino acid and fatty acid catabolism occurs.

(b) Under conditions of hyperinsulinism, little circulating fuel is available to meet the requirements for ATP production, so the functions of energy-demanding tissues, such as brain and muscle, are compromised. This accounts for the shaking and trembling, weakness and fatigue. Excess insulin affects the mechanisms that regulate the autonomic nervous system, which in turn controls sweating. Finally, brain damage can result from lack of glucose because glucose is the main source of fuel for the brain.

13. Thermogenesis Caused by Thyroid Hormones Thyroid hormones are intimately involved in regulating the basal metabolic rate. Liver tissue of animals given excess thyroxine shows an increased rate of O_2 consumption and increased heat output (thermogenesis), but the ATP concentration in the tissue is normal. Different explanations have been offered for the thermogenic effect of thyroxine. One is that excess thryroxine causes uncoupling of oxidative phosphorylation in mitochondria. How could such an effect account for the observations? Another explanation suggests that the thermogenesis is due to an increased rate of ATP utilization by the thyroxine-stimulated tissue. Is this a reasonable explanation? Why?

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Answer The observations are consistent with the thesis that thyroxine acts as an uncoupler of oxidative phosphorylation. Uncouplers lower the P/O ratio of tissues, and thus the tissue must increase respiration to meet the normal ATP demands. The observed thermogenesis could also be due to the increased rate of ATP utilization by the thyroxine-stimulated tissue, as the increased ATP demands are met by increased oxidative phosphorylation and thus respiration.

14. Function of Prohormones What are the possible advantages in the synthesis of hormones as prohormones?

Answer Because prohormones are inactive, they can be stored in quantity in secretory granules. Rapid activation is achieved by enzymatic cleavage in response to an appropriate signal.

15. Sources of Glucose during Starvation The typical human adult uses about 160 g of glucose per day, 120 g of which is used by the brain. The available reserve of glucose (~20 g of circulating glucose and ~190 g of glycogen) is adequate for about one day. After the reserve has been depleted during starvation, how would the body obtain more glucose?

Answer In animals, glucose can be synthesized from a variety of precursors by gluconeogenesis (Fig. 14–15). In humans, the principal precursors (obtained from storage depots) are glycerol from triacylglycerols and glucogenic amino acids from proteins. Oxaloacetate formed from CO_2 and pyruvate by pyruvate carboxykinase is also a source of glucose.

16. Parabiotic *ob/ob* mice By careful surgery, researchers can connect the circulatory systems of two mice so that the same blood circulates through both animals. In these **parabiotic** mice, products released into the blood by one animal reach the other animal via the shared circulation. Both animals are free to eat independently. If an *ob/ob* mouse (both copies of the *OB* gene are defective) and a normal *OB/OB* mouse (two good copies of the *OB* gene) were made parabiotic, what would happen to the weight of each mouse?

Answer The mutant (*ob/ob*) mouse is obese because it does not produce functional leptin, the signal to eat less and be more physically active. When the circulatory system of this mouse is joined with that of the normal mouse, leptin produced in the normal mouse reaches the hypothalamus of the obese mouse, triggering the "eat less" response; the obese mouse therefore loses weight. The normal *OB/OB* mouse continues to have enough leptin in its blood to maintain the signal to eat moderately, and it retains its normal body weight.

17. Calculation of Body Mass Index A portly biochemistry professor weighs 260 lb (118 kg) and is 5 feet 8 inches (173 cm) tall. What is his body mass index? How much weight would he have to lose to bring his body mass index down to 25 (normal)?

Answer BMI = [weight in kg]/[(height in m)^2]. The professor's current BMI = $118/(1.73)^2 = 39.5$. To achieve a BMI of 25, his weight would need to be $25(1.73)^2 = 74.8$ kg. He must lose current weight (118 kg) − desired weight (74.8 kg) = 43.2 kg (95.0 lb).

18. Insulin Secretion Predict the effects on insulin secretion by pancreatic β cells of exposure to the potassium ionophore valinomycin (p. 404). Explain your prediction.

Answer Valinomycin is a K^+ ionophore; it makes the membrane permeable to K^+. Valinomycin has the same effect as opening the K^+ channel, allowing exit of K^+ and consequent hyperpolarization of the membrane. This keeps the voltage-gated Ca^{2+} channel closed and reduces secretion of insulin from the cells. See Figure 23–28, p. 924.

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19. Effects of a Deleted Insulin Receptor A strain of mice specifically lacking the insulin receptor of liver is found to have mild fasting hyperglycemia (blood glucose = 132 mg/dL, vs. 101 mg/dL in controls) and a more striking hyperglycemia in the fed state (glucose = 363 mg/dL, vs. 135 mg/dL in controls). The mice have higher than normal levels of glucose 6-phosphatase in the liver and elevated levels of insulin in the blood. Explain these observations.

Answer In these mice the hepatocytes lack the insulin receptor and so are insensitive to the blood insulin level. The cells continue to have high levels of glucose 6-phosphatase and high rates of gluconeogenesis, thus increasing blood glucose both during a fast and after a glucose-containing meal. The elevated blood glucose concentration triggers insulin release from pancreatic β cells. This accounts for the observed high level of insulin in the blood.



20. Decisions on Drug Safety The drug Avandia (rosiglitazone) is effective in lowering blood glucose in patients with type 2 diabetes, but also seems to carry an increased risk of heart attack. If it were your responsibility to decide whether this drug should remain on the market (labeled with suitable warnings of its side effects) or should be withdrawn, what factors would you weigh in making your decision?

Answer Some things to consider: What is the frequency of heart attack attributable to the drug? How does this frequency compare with the number of individuals spared the long-term consequences of type 2 diabetes? Are other, equally effective treatment options, with fewer adverse effects, available?



21. Type 2 Diabetes Medication The drugs acarbose (Precose) and miglitol (Glyset), used in the treatment of type 2 diabetes mellitus, inhibit α-glucosidases in the brush border of the small intestine. These enzymes degrade oligosaccharides derived from glycogen or starch to monosaccharides. Suggest a possible mechanism for the salutary effect of these drugs on individuals with diabetes. What side effects, if any, would you expect from these drugs. Why? (Hint: Review lactose intolerance, p. 545).

Answer Without intestinal glucosidase activity, absorption of glucose from dietary glycogen and starch is reduced, blunting the usual rise in blood glucose after the meal. The undigested oligosaccharides are fermented by bacteria in the large intestine, and the gases released cause intestinal discomfort.

Data Analysis Problem

22. Cloning the Pancreatic β-Cell Sulfonylurea Receptor Glyburide, a member of the sulfonylurea family of drugs shown on p. 925, is used to treat type 2 diabetes. It binds to and closes the ATP-gated K^+ channel shown in Figures 23–28 and 23–29.

(a) Given the mechanism shown in Figure 23–28, would treatment with glyburide result in increased or decreased insulin secretion by pancreatic β cells? Explain your reasoning.

(b) How does treatment with glyburide help reduce the symptoms of type 2 diabetes?

(c) Would you expect glyburide to be useful for treating type 1 diabetes? Why or why not?

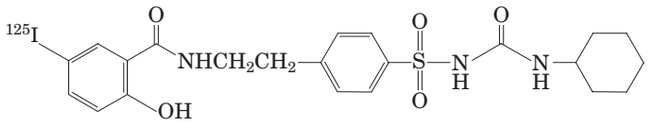
Aguilar-Bryan and coauthors (1995) cloned the gene for the sulfonylurea receptor (SUR) portion of the ATP-gated K^+ channel from hamsters. The research team went to great lengths to ensure that the gene they cloned was in fact the SUR-encoding gene. Here we explore how it is possible for researchers to demonstrate that they have actually cloned the gene of interest rather than another gene.

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The first step was to obtain pure SUR protein. As was already known, drugs such as glyburide bind SUR with very high affinity ($K_d < 10$ nM), and SUR has a molecular weight of 140 to 170 kDa. Aguilar-Bryan and coworkers made use of the high-affinity glyburide binding to tag the SUR protein with a radioactive label that would serve as a marker to purify the protein from a cell extract. First, they made a radiolabeled derivative of glyburide, using radioactive iodine (^{125}I):



[^{125}I]5-Iodo-2-hydroxyglyburide

(d) In preliminary studies, the ^{125}I-labeled glyburide derivative (hereafter, [^{125}I]glyburide) was shown to have the same K_d and binding characteristics as unaltered glyburide. Why was it necessary to demonstrate this (what alternative possibilities did it rule out)?

Even though [^{125}I]glyburide bound to SUR with high affinity, a significant amount of the labeled drug would probably dissociate from the SUR protein during purification. To prevent this, [^{125}I]glyburide had to be covalently cross-linked to SUR. There are many methods for covalent cross-linking; Aguilar-Bryan and coworkers used UV light. When aromatic molecules are exposed to short-wave UV, they enter an excited state and readily form covalent bonds with nearby molecules. By cross-linking the radiolabeled glyburide to the SUR protein, the researchers could simply track the ^{125}I radioactivity to follow SUR through the purification procedure.

Aguilar-Bryan and colleagues treated hamster HIT cells (which express SUR) with [^{125}I]glyburide and UV light, purified the ^{125}I-labeled 140 kDa protein, and sequenced its amino-terminal 25 amino acid segment; they found the sequence PLAFCGTENHSAAYRVDQGVLNNGC. The investigators then generated antibodies that bound to two short peptides in this sequence, one that bound to PLAFCGTE and the other to HSAAYRVDQGV, and showed that these antibodies bound the purified ^{125}I-labeled 140 kDa protein.

(e) Why was it necessary to include this antibody-binding step?

Next, the researchers designed PCR primers based on the sequences above, and cloned a gene from a hamster cDNA library that encoded a protein that included these sequences (see Chapter 9 on biotechnology methods). The cloned putative *SUR* cDNA hybridized to an mRNA of the appropriate length that was present in cells known to contain SUR. The putative *SUR* cDNA did not hybridize to any mRNA fraction of the mRNAs isolated from hepatocytes, which do not express SUR.

(f) Why was it necessary to include this putative *SUR* cDNA–mRNA hybridization step?

Finally, the cloned gene was inserted into and expressed in COS cells, which do not normally express the *SUR* gene. The investigators mixed these cells with [^{125}I]glyburide with or without a large excess of unlabeled glyburide, exposed the cells to UV light, and measured the radioactivity of the 140 kDa protein produced. Their results are shown in the table.

Experiment	Cell type	Added putative SUR cDNA?	Added excess unlabeled glyburide?	^{125}I label in 140 kDa protein
1	HIT	No	No	+++
2	HIT	No	Yes	−
3	COS	No	No	−
4	COS	Yes	No	+++
5	COS	Yes	Yes	−

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(g) Why was no ^{125}I-labeled 140 kDa protein found in experiment 2?

(h) How would you use the information in the table to argue that the cDNA encoded SUR?

(i) What other information would you want to collect to be more confident that you had cloned the *SUR* gene?

Answer

(a) Closing the ATP-gated K^+ channel would depolarize the membrane, leading to increased insulin release.

(b) Type 2 diabetes results from decreased sensitivity to insulin, not a deficit of insulin production; increasing circulating insulin levels will reduce the symptoms associated with this disease.

(c) Individuals with type 1 diabetes have deficient pancreatic β cells, so glyburide will have no beneficial effect.

(d) Iodine, like chlorine (the atom it replaces in the labeled glyburide), is a halogen, but it is a larger atom and has slightly different chemical properties. It is possible that the iodinated glyburide would not bind to SUR. If it bound to another molecule instead, the experiment would result in cloning of the gene for this other, incorrect protein.

(e) Although a protein has been "purified," the "purified" preparation might be a mixture of several proteins that co-purify under those experimental conditions. In this case, the amino acid sequence could be that of a protein that co-purifies with SUR. Using antibody binding to show that the peptide sequences are present in SUR excludes this possibility.

(f) Although the cloned gene does encode the 25 amino acid sequence found in SUR, it could be a gene that, coincidentally, encodes the same sequence in another protein. In this case, this other gene would most likely be expressed in different cells than the *SUR* gene. The mRNA hybridization results are consistent with the putative *SUR* cDNA actually encoding SUR.

(g) The excess unlabeled glyburide competes with labeled glyburide for the binding site on SUR. As a result, there is significantly less binding of labeled glyburide, so little or no radioactivity is detected in the 140 kDa protein.

(h) In the absence of excess unlabeled glyburide, labeled 140 kDa protein is found only in the presence of the putative *SUR* cDNA. Excess unlabeled glyburide competes with the labeled glyburide, and no ^{125}I-labeled 140 kDa protein is detected. This shows that the cDNA produces a glyburide-binding protein of the same molecular weight as SUR—strong evidence that the cloned gene encodes the SUR protein.

(i) Several additional steps are possible, such as: (1) Express the putative *SUR* cDNA in CHO (Chinese hamster ovary) cells and show that the transformed cells have ATP-gated K^+ channel activity. (2) Show that HIT cells with mutations in the putative *SUR* gene lack ATP-gated K^+ channel activity. (3) Show that experimental animals or human patients with mutations in the putative *SUR* gene are unable to secrete insulin.

Reference

Aguilar-Bryan, L., Nichols, C.G., Wechsler, S.W., Clement, J.P. IV, Boyd, A.E. III, González, G., Herrera-Sosa, H., Nguy, K., Bryan, J., & Nelson, D.A. (1995) Cloning of the β cell high-affinity sulfonylurea receptor: a regulator of insulin secretion. *Science* **268,** 423–426.

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